



Multiple pregnancy

Dr. Massooma Mahmood Mohammed
GBCOG,MB.CH.B
College of medicine, Sulaimania university
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Definition

- * In general terms, multiple pregnancies consist of two or more fetuses.
- * Twins make up the vast majority (nearly 99 per cent) of multiple gestations.
- * Pregnancies with three or more fetuses are referred to as 'higher multiples'.
- * twins currently account for approximately 1.5 per cent of all pregnancies
- * Rates of multiple pregnancies constitute approximately 3% of live births.
- * The incidence of monozygotic or identical twins is generally accepted to be constant at 1 in 250. It is not influenced by race, family history or parity.

Risk Factors

Risk factors for multiple gestations include (dizygotic twins)

- * Previous multiple pregnancy
- * Family history (maternal side)
- * Increasing maternal age
- * Racial origin (more common in women of West African ancestry, less common in those of Japanese ancestry)
- * Assisted conception

Aetiology

- ❖ Dizygotic twins arise spontaneously from the release of two eggs at ovulation.

This tendency to release more than one egg can be familial or racial in origin and increases with maternal age.

- ❖ Ovulation induction treatments may also cause the release of more than one egg.

- ❖ With assisted conception techniques such as IVF, two or more embryos fertilized in the laboratory may be replaced in the uterus, although single embryo transfer is becoming more frequent.

classification

The classification of multiple pregnancy is based on:

- * number of fetuses: twins, triplets, quadruplets, etc.
- number of fertilized eggs: zygosity
- number of placentae: chorionicity
- number of amniotic cavities: amnionicity

Non- identical Twins



Non-identical or fraternal twins are dizygotic, having resulted from the fertilization of two separate eggs.

- * they always have two functionally separate placentae (dichorionic).
- * always have separate amniotic cavities (diamniotic)
- * and the two cavities are separated by a thick three-layer membrane(fused amnion in the middle with chorion on either side).
- * The fetuses can be either same-sex or different sex pairings.



Health & Parenting

Identical Twins



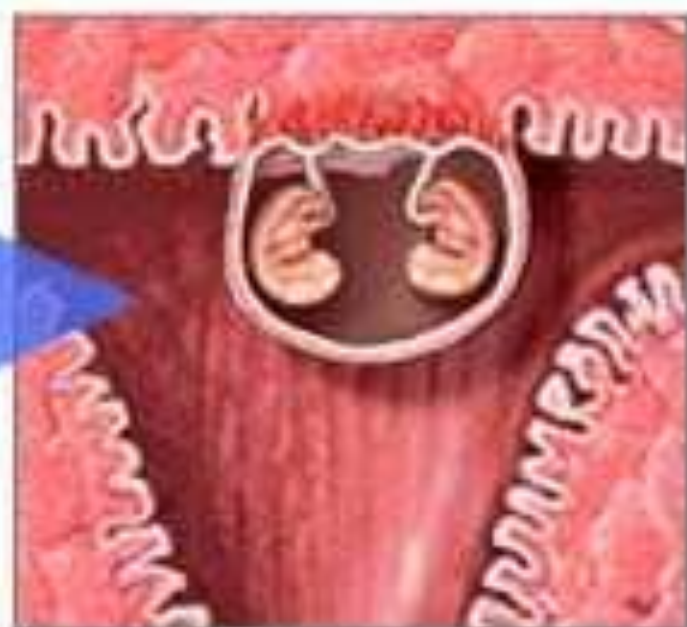
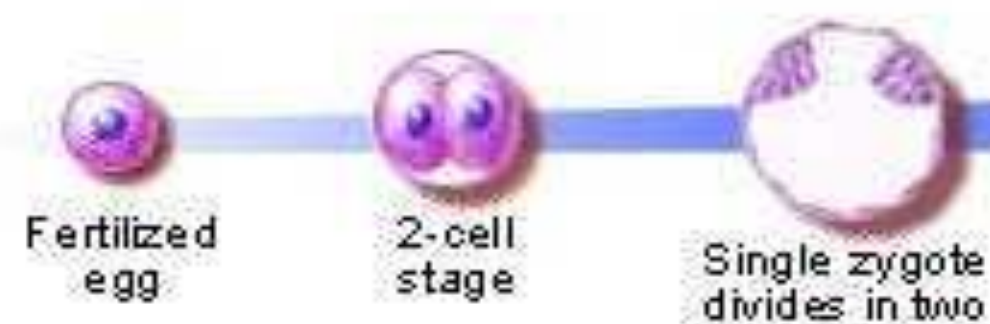
Identical twins are monozygotic – they arise from fertilization of a single egg and are always same-sex pairings.

They may share a single placenta (monochorionic) or have one each (dichorionic).

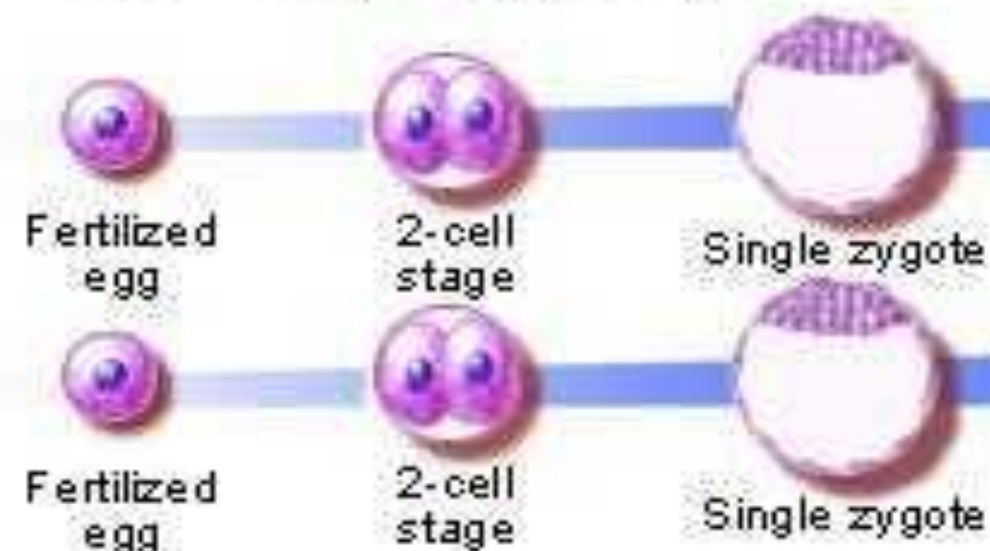
The vast majority of monochorionic twins have two amniotic cavities (diamniotic) but the dividing membrane is thin, as it consists of a single layer of amnion alone.

Monochorionic twins may occasionally share a single sac (monoamniotic)

Identical (monozygotic) twins

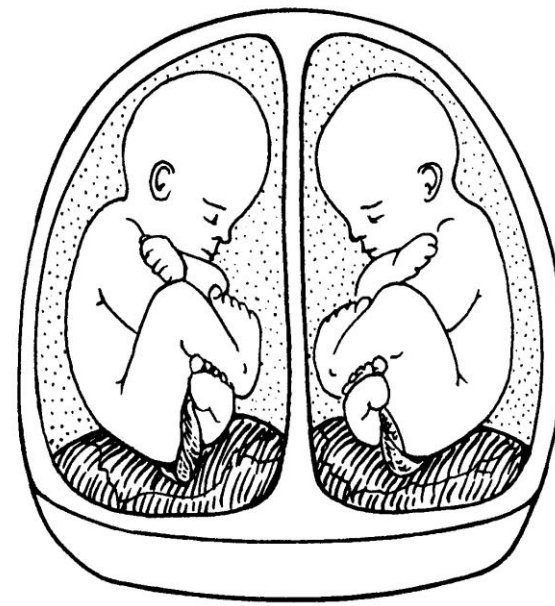


Fraternal (dizygotic) twins

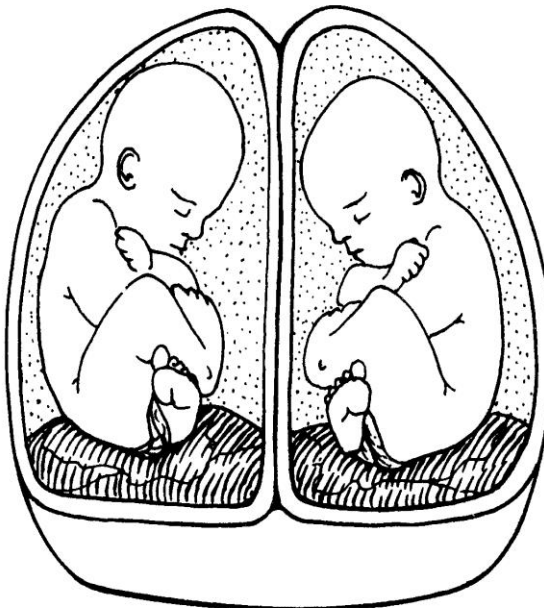




Monochorionic/Monoamniotic



Monochorionic/Diamniotic



Dichorionic/Diamniotic
(Fused Placenta)



Dichorionic/Diamniotic
(Separate Placenta)

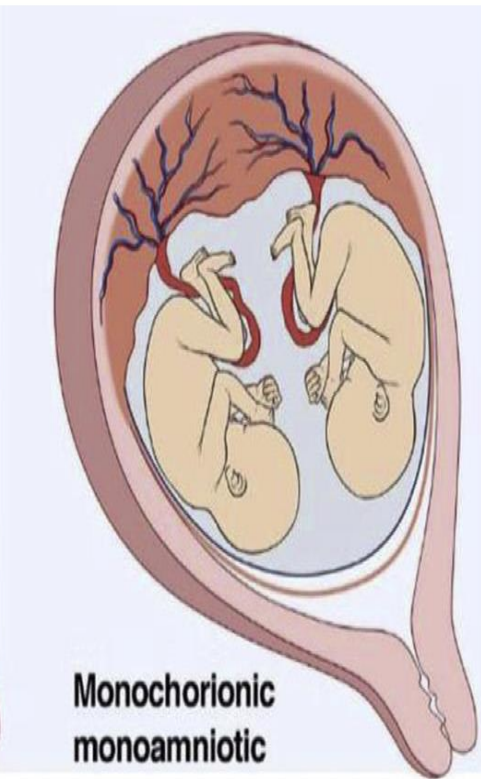
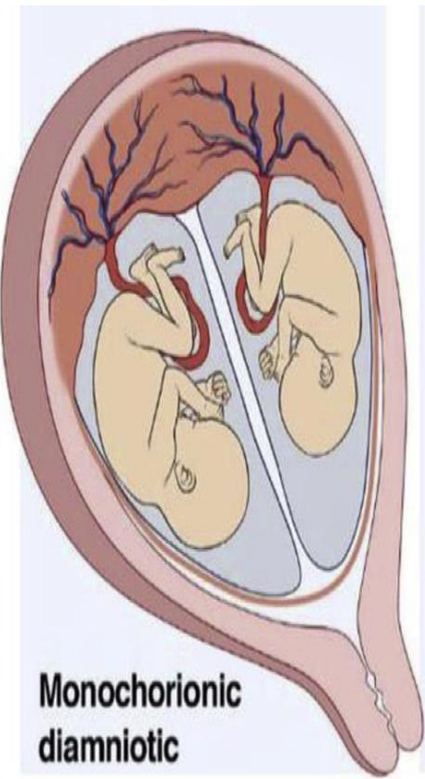
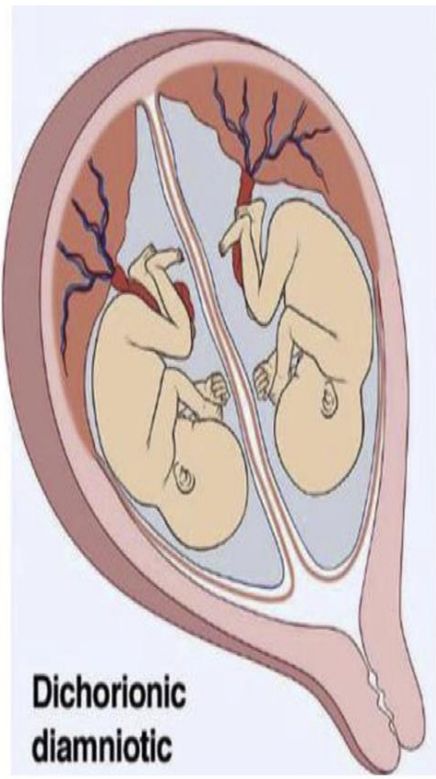
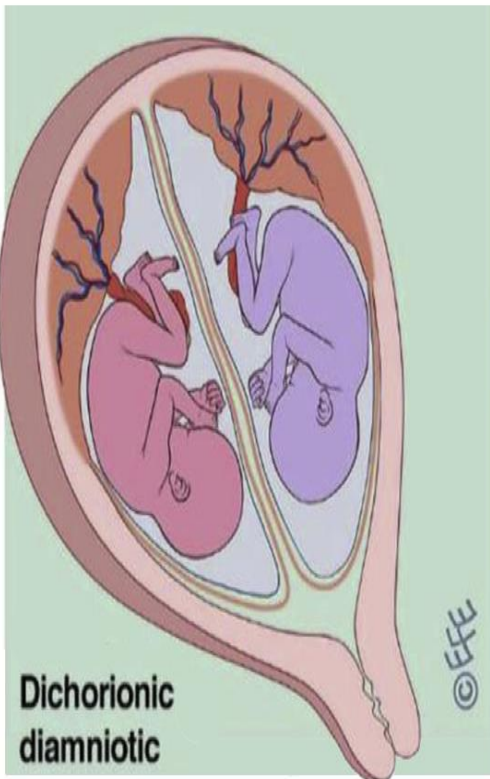


Figure A

Seperate placentas
Two chorions
Two amnions

19% of MZ twins
58% of DZ twins

Figure B

"Fused" placentas
Two chorions
Two amnions

13% of MZ twins
42% of DZ twins

Figure C

Single placentas
One chorion
Two amnions

64% of MZ twins

Figure D

Single placentas
One chorions
One amnion

4% of MZ twins

Identical Twins

- ❖ Monozygotic twins arise from a single fertilized ovum that splits into two identical structures.
- ❖ The type of monozygotic twins depends on how long after conception the split occurs.
- ❖ If splitting is delayed beyond day 12, the embryonic disc has also formed, and conjoined, or 'Siamese' twins will result.

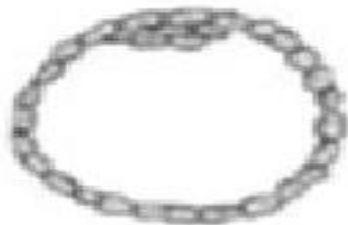
**Cleavage
after
fertilization**

Days 1-4



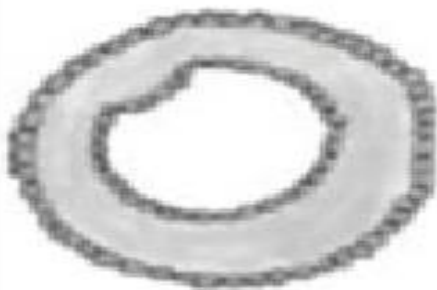
Morula

Days 5-6



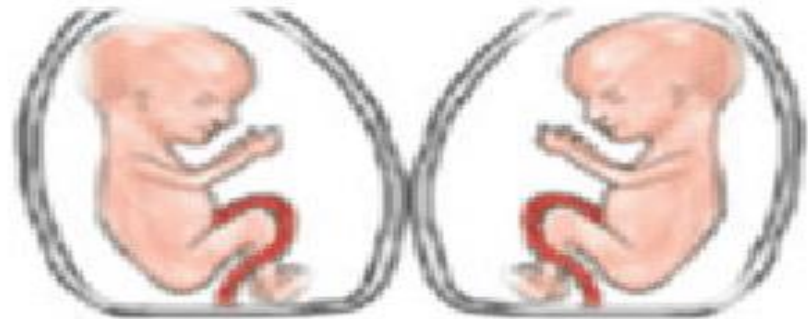
Blastocyst

Days 8-12

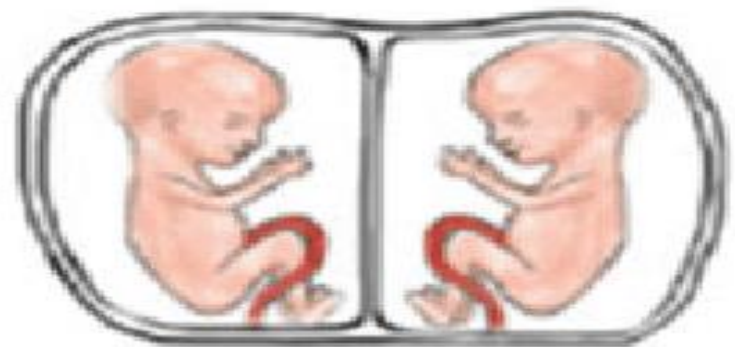


**Implanting
Blastocyst**

Dichorionic-Diamniotic

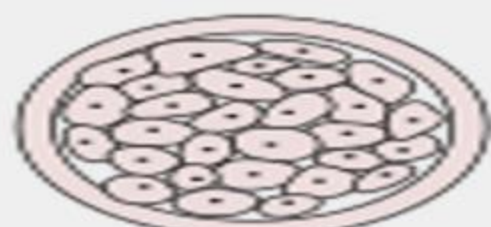


Monochorionic-Diamniotic



**Monochorionic
Monoamniotic**





Morula

Cleavage
Days 1-3



Dichorionic/Diamniotic

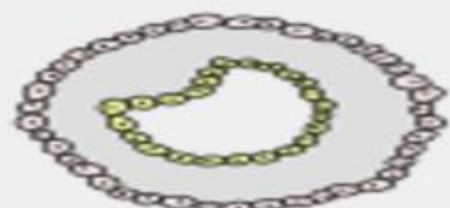


Blastocyst

Cleavage
Days 4-8



Monochorionic/Diamniotic

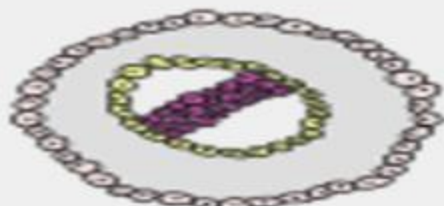


Implanted
Blastocyst

Cleavage
Days 8-13



Monochorionic/Monoamniotic



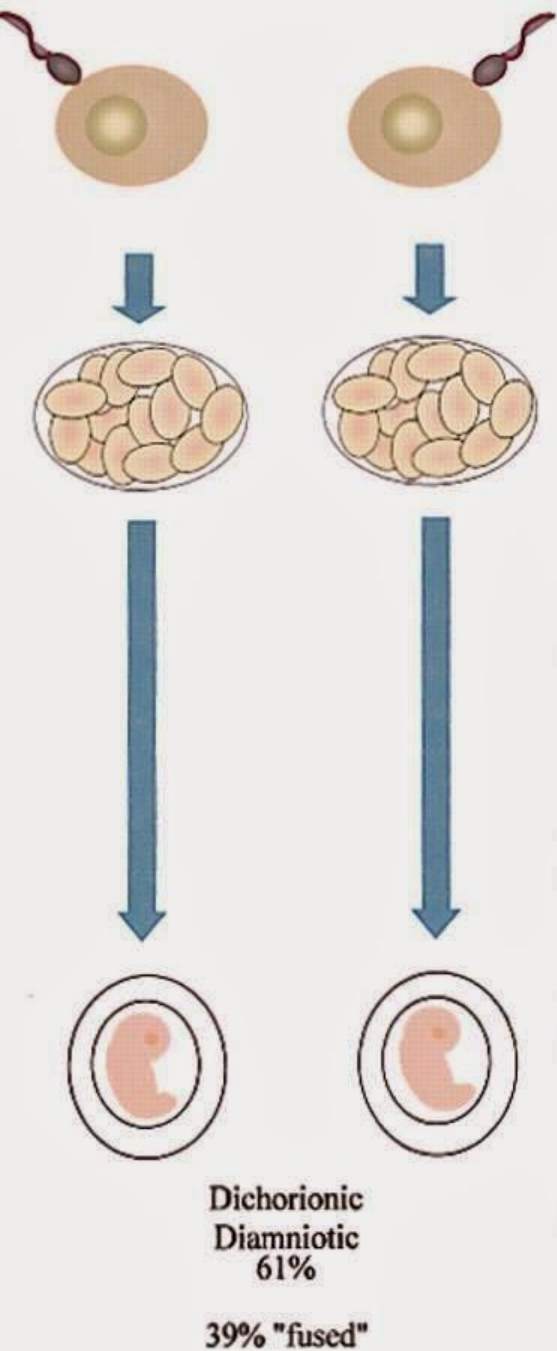
Formed
Embryonic Disc

Cleavage
Days 13-15

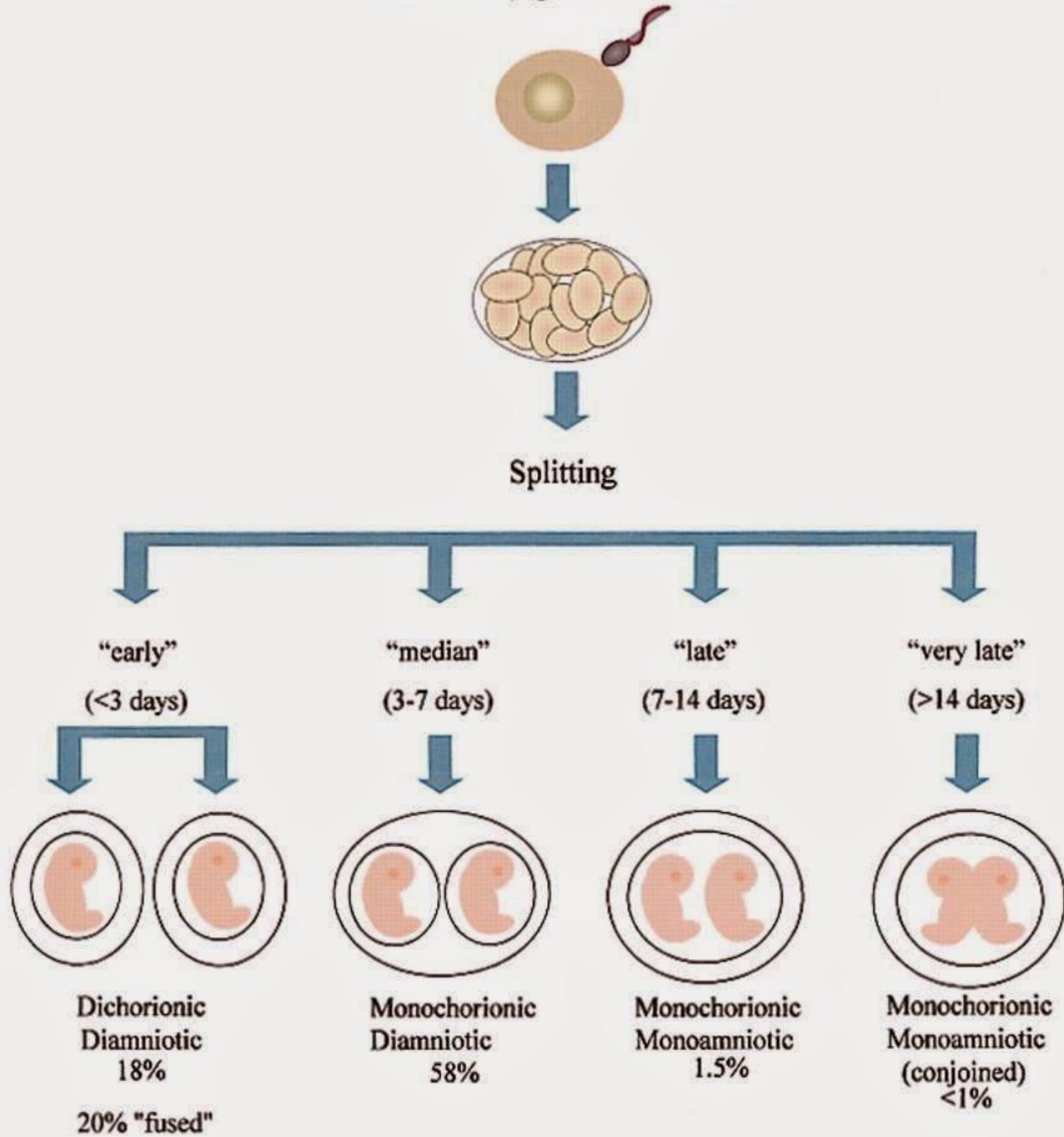


Conjoined Twins

Dizygotic Twins



Monozygotic Twins



Complications relevant to twin pregnancy

- * All the physiological changes of pregnancy (increased cardiac output, volume expansion, relative hemodilution, diaphragmatic splinting, weight gain,) are exaggerated in multiple gestations.
- * **Miscarriage** and severe preterm delivery
- * Spontaneous preterm delivery is an ever-present risk in any twin pregnancy the average gestation at delivery is 37 weeks. about half of all twins deliver preterm.
- * Increased risk for pre-eclampsia, hypertension, gestational diabetes and FGR, APH, PPH and thromboembolic disease.
- * Monochorionic monoamniotic (MCMA) twins have increased risk of congenital anomalies including neural tube defects and abdominal wall and urinary tract malformations.

Complications relevant to twin pregnancy

- * **Twin-to-twin transfusion syndrome** (TTTS) and, more rarely, twin anaemia–polycythaemia sequence (TAPS)
- * The **death** of one of a monochorionic twin pair may result in either death or handicap of the co-twin because of acute hypotension secondary to placental vascular anastomoses between the two circulations.
- * Compared to singletons, twin pregnancies carry at least twice the risk of the birth of a baby with **an anomaly**.
- * The overall infant **mortality rate** for twins is around 5.5 times higher than for singletons.
- * Increased risk of chromosomal abnormalities

Death of one fetus in a twin pregnancy



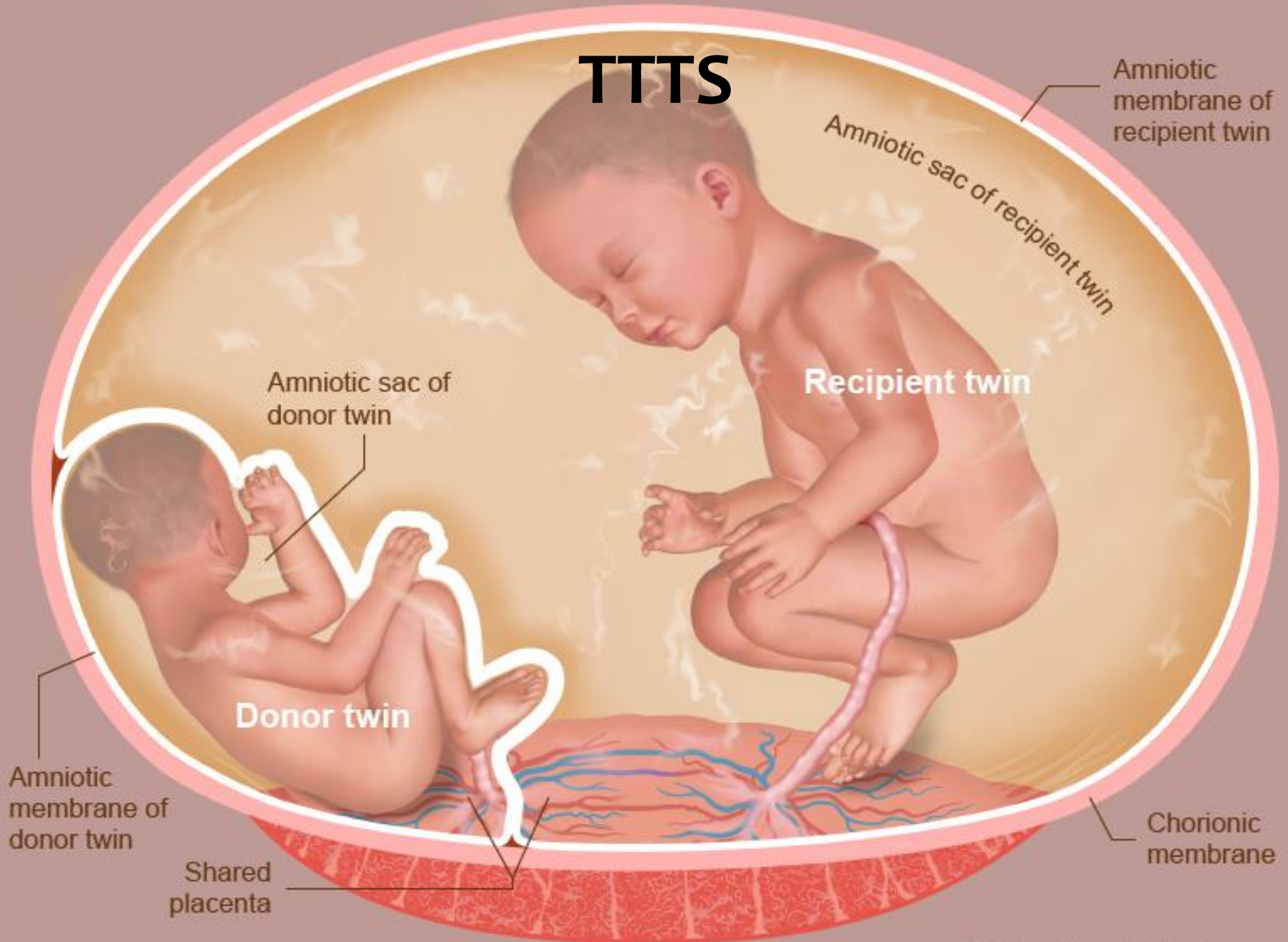
Complications unique to monochorionic twin pregnancies

- * **TTTS:** involving the development of abnormal unbalanced vascular anastomoses leading to the development of TTTS. Although vascular connections are found in nearly all monochorionic twins, approximately 10% of monochorionic diamniotic pregnancies and 5% of monoamniotic pregnancies will subsequently develop TTTS.
- **TTTS** is diagnosed based on the following ultrasound criteria:
 - Single placental mass.
 - * Concordant gender.
 - * Oligohydramnios with maximum vertical pool (MVP) less than 2 cm in one sac and polyhydramnios in the other sac (MVP >8 cm).
 - * Discordant bladder appearances.
 - * Haemodynamic and cardiac compromise

Complications unique to monochorionic twin pregnancies

- * **TAPS** is a rarer chronic form of TTTS in which a large inter-twin haemoglobin difference occurs but the oligohydramnios polyhydramnios sequence that is observed with TTTS is not seen.
- * Severe polycythaemia can occur, leading to fetal and placental thrombosis and hydrops fetalis in the anaemic twin.
- * **Treatment of TTTS:(management options include):**
- * expectant management; amnioreduction; septostomy; selective feticide, fetoscopic laser ablation of vascular anastomoses.
- * Fetoscopic laser ablation is now generally considered the definitive treatment for severe TTTS between 16 and 26 weeks' gestation. Above 26 weeks, delivery may be considered.

TTTS



Fetoscope

TTTS

Amniotic
membrane of
recipient twin

Amniotic sac of recipient twin

Recipient twin

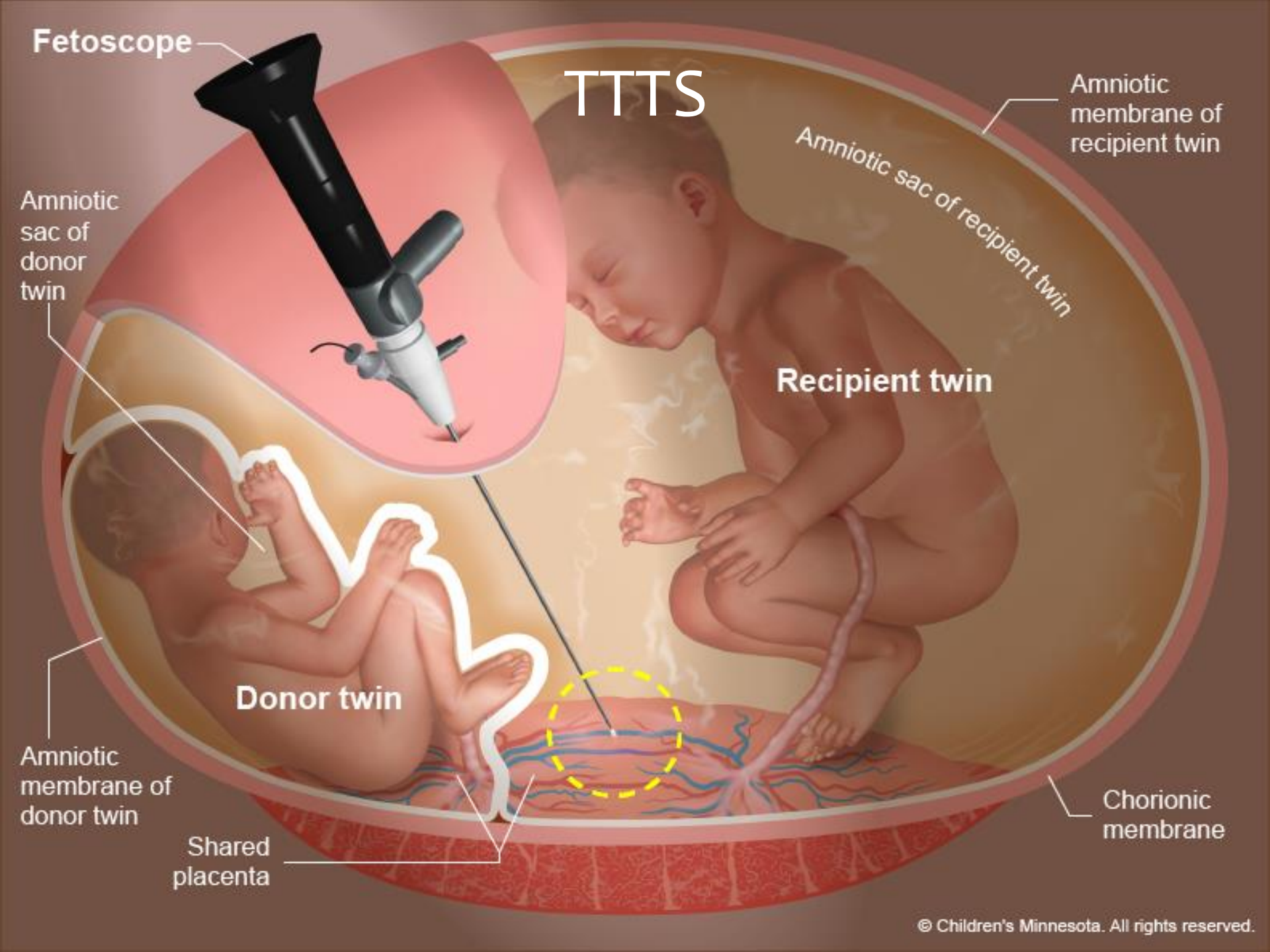
Amniotic
sac of
donor
twin

Donor twin

Amniotic
membrane of
donor twin

Shared
placenta

Chorionic
membrane



Antenatal care

- * women with uncomplicated dichorionic twin pregnancies be offered at least eight antenatal appointments and at least nine antenatal appointments for uncomplicated monochorionic diamniotic twin pregnancies). Routine antenatal care for all women involves screening for hypertension and gestational diabetes.
- * Routine supplementation in multiple pregnancies. Minor symptoms of pregnancy are more common, but management is again unchanged compared to singletons.
- * Screening in multiple pregnancy: The purpose of 1st trimester scan in multiple pregnancy is threefold: 1-To accurately estimate gestational age 2-To determine chorionicity 3-To screen for Down's syndrome.

Determination of chorionicity

Determination of chorionicity is critical to 'good' management, and this is done most reliably by ultrasound in the late first trimester.

- * In dichorionic twins, there is a V-shaped extension of placental tissue into the base of the inter-twin membrane, referred to as the 'lambda' or 'twin-peak' sign.
- * In monochorionic twins, this sign is absent and the inter-twin membrane joins the uterine wall in a T shape

The twin peak sign (lambda (λ) sign)





Ultrasound appearance of monochorionic (a)
and dichorionic

Absent twin peak

Screening for fetal abnormalities

- * Screening for trisomy 21 using maternal serum biochemistry .
- * The measurement of nuchal translucency at 12 weeks gestation allows each fetus to have an individualized assessment of risk.
- * Both amniocentesis and chorion villus sampling (CVS) can be performed in twin pregnancies, but in dichorionic pregnancies, it is essential that both fetuses are sampled.
- * Screening for fetal structural anomalies is done using second trimester ultrasound, with optimal detection rates seen at 20 weeks gestation.

Timing of delivery

- * **uncomplicated dichorionic diamniotic twin pregnancy:**
- * planned birth from 37⁺⁰ weeks **and** continuing the pregnancy beyond 37⁺⁶ weeks increases the risk of fetal death.
- * **uncomplicated monochorionic diamniotic twin pregnancy:**
- * planned birth from 36⁺⁰ weeks **and** continuing the pregnancy beyond 36⁺⁶ weeks increases the risk of fetal death.
- * **uncomplicated monochorionic monoamniotic twin pregnancy :** planned birth between 32⁺⁰ and 33⁺⁶ weeks does not appear to be associated with an increased risk of serious neonatal adverse outcomes **and** continuing the pregnancy beyond 33⁺⁶ weeks increases the risk of fetal death.

Delivery and Intrapartum management

- * A delivery plan should be discussed and made with the patient, ideally throughout pregnancy, with a clear delivery plan discussed and documented early in the third trimester.
- * This should begin long before labor, with antenatal education and an intrapartum care plan.
- * On admission in labour, obtain IV access, blood grouped and saved, monitor fetal heart rates separately and check position of lead fetus:
 - In 60% case both fetuses present as cephalic
 - In 20% cases 1st twin cephalic, 2nd breech
 - In 10% cases breech/cephalic
 - In 10% cases breech/ breech
- *

Intrapartum management

- * **The ideal criteria for a twin labour are:**

- Spontaneous onset
- Cephalic presentation of twin 1
- Twin 1 larger than twin 2
- Dichorionic pregnancy

Complications in labor are more common with twin gestations.

These include:

- * premature birth
- * Abnormal presentations
- * prolapsed cord
- * premature separation of the placenta
- * postpartum haemorrhage.



45%

Vertex and Vertex



37%

Vertex and Breech



10%

Breech and Breech



5%

Vertex and Transverse



2%

Breech and Transverse



0.5%

Transverse and Transverse

Intrapartum management

- * twin CTG machine should be used for fetal monitoring and a portable ultrasound machine should be available during the delivery.
- * A standard oxytocin solution for augmentation should be prepared, run through an intravenous giving-set and clearly labelled 'for augmentation', for use for delivery of the second twin, if required.
- * high-dose oxytocin infusion should also be available for the management of postpartum haemorrhage.
- * Analgesia, ideally in the form of an early epidural, to allow for internal podalic version (if needed) for twin 2.

Vaginal delivery of vertex–vertex

- * Delivery of the first twin is undertaken in the usual manner and thereafter the majority of second twins will be delivered within 15 minutes. However, there is no urgency to deliver the second twin within a set time period, providing both mother and baby remain well.
- * After the delivery of the first twin, abdominal palpation should be performed to assess the lie of the second twin. If the lie is longitudinal with a cephalic presentation, one should wait until the head is descending and then perform amniotomy with a contraction.
- * If contractions do not ensue within 5–10 minutes after delivery of the first twin, an oxytocin infusion should be started.
- * The indications for instrumental delivery of the second twin are as for singletons .

Delivery of vertex–non-vertex

- * If the second twin is non-vertex, which occurs in about 40 per cent of twins, numerous studies have shown that vaginal delivery can be safely considered.
- * If the second twin is a breech, the membranes can be ruptured once the breech is fixed in the birth canal
- * A total breech extraction may be performed if fetal distress occurs or if a footling breech is encountered, but this requires considerable expertise.
- * Where the fetus is transverse, external cephalic version can be successful in more than 70 per cent of cases.
- * If external cephalic version is unsuccessful, and assuming that the operator is experienced, an internal podalic version can be undertaken

Internal podalic version



Non-vertex first twin

When the first twin presents as a breech, clinicians usually recommend delivery by elective Caesarean section.

